Pixel-Based Optical Navigation in Cislunar Space

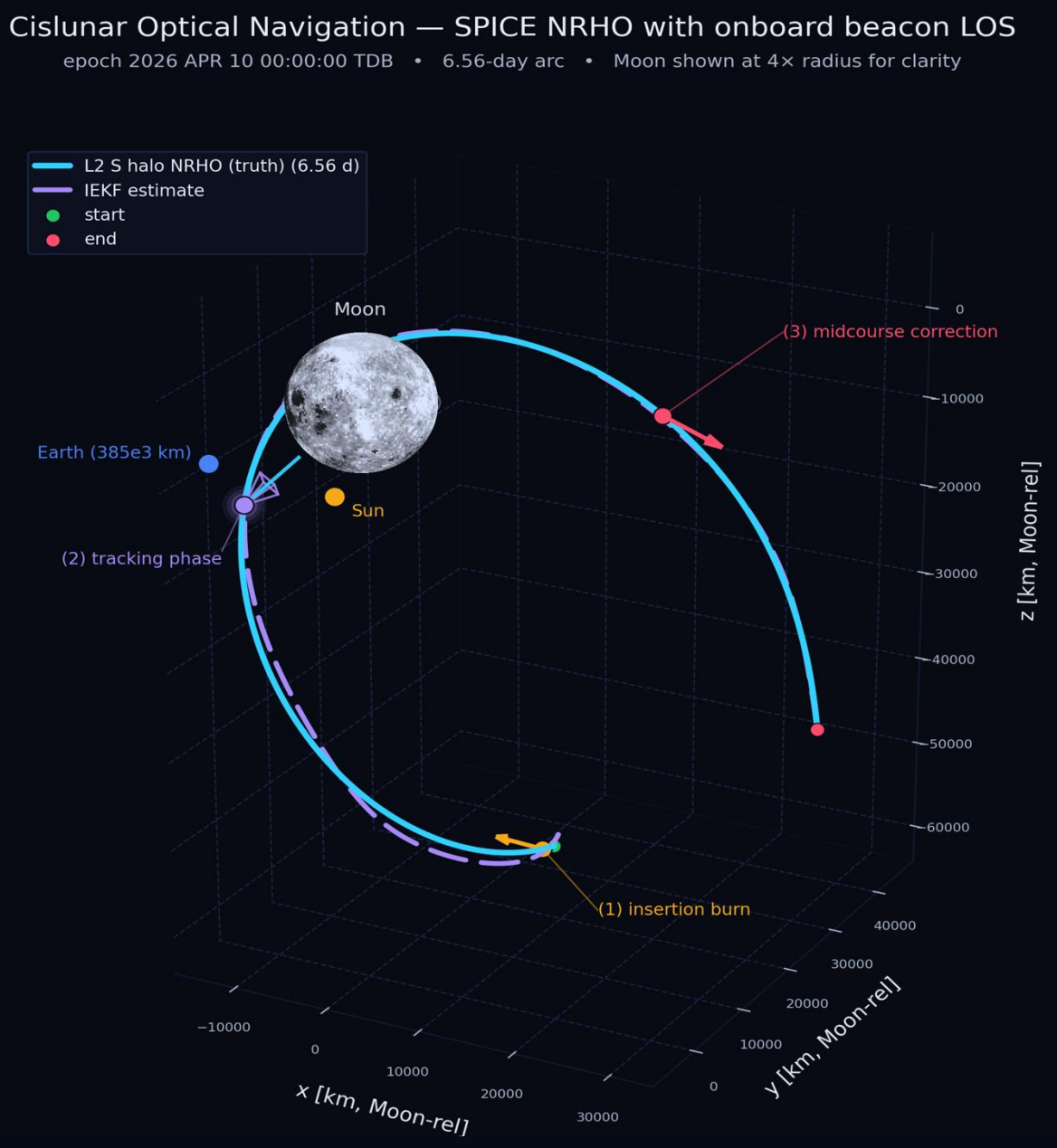
Bearing-Only State Estimation

Active Tracking

Midcourse Correction

Enabling autonomous navigation beyond Earth orbit using onboard vision

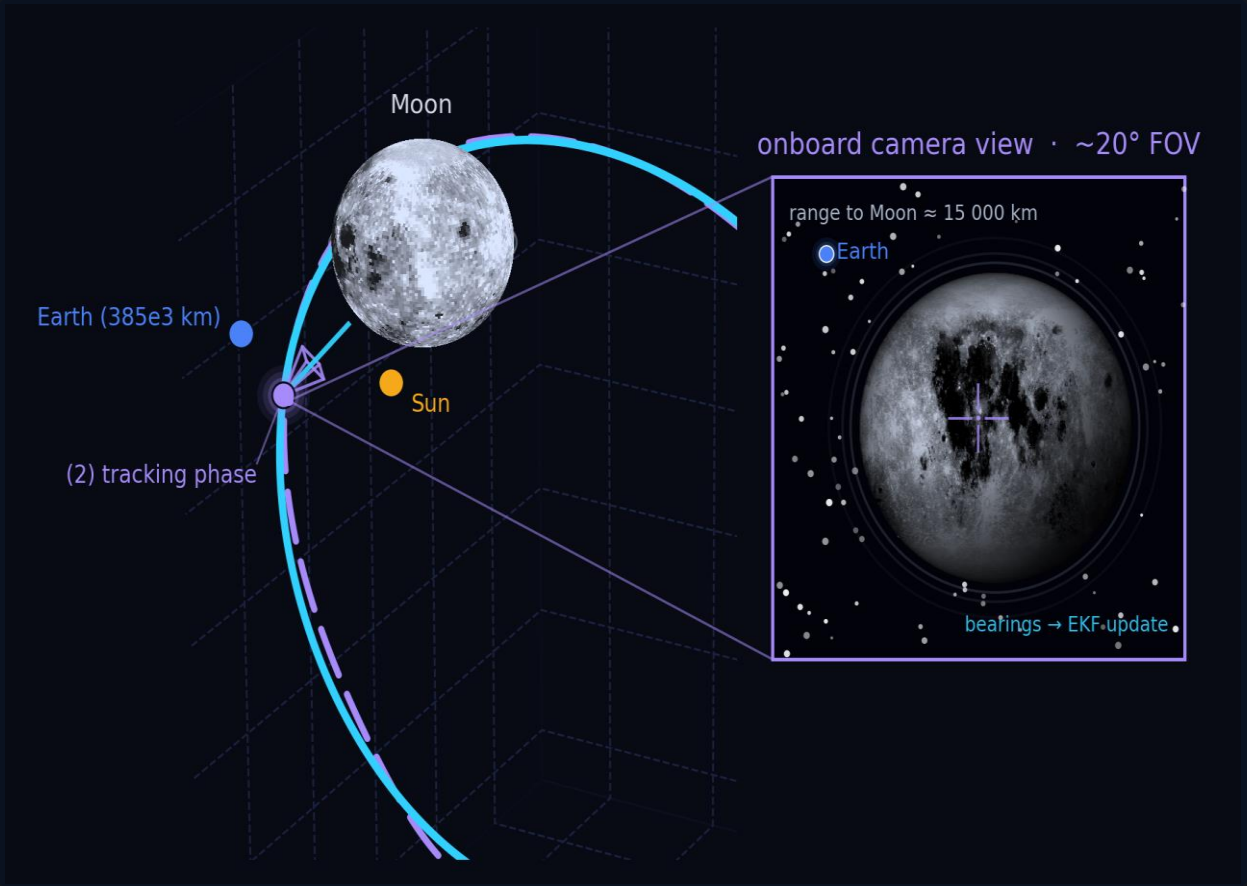
Phases 00-07 Demo for ASTE 581 Final



V. Krishnakumar (GNC/CV Engineer)

Why use a Camera for Cislunar Navigation?

In cislunar space, navigation cannot rely on strong, continuous GPS-like positioning support, so onboard vision becomes an attractive source of autonomous navigation information



Navigation Challenge

- Spacecraft may not have continuous access to strong GPS-like navigation support
- Ground-based tracking remains valuable, but onboard autonomy becomes increasingly important

Why a camera?

- Passive, lightweight, and power-efficient
- Provides continuous geometric information whenever the target is visible
- Naturally supports landmark tracking and horizon sensing

Tradeoff

- Camera does not directly measure range (ρ)
- Provides image-plane info (u,v) that becomes a line-of-sight direction
- Missing direction must be recovered from dynamics and repeated observations

Project idea: use onboard vision to replace or supplement weak navigation observables in cislunar space

Cislunar Optical Navigation — Software Pipeline

SPICE-backed IEKF with autonomous camera pointing (Phase 7)

This talk focuses on Phases 02–07 · Phase 08 closes as a future-facing extension

Truth · Measurements · Estimation · Guidance · Control

ZONE 1 · Dynamics / Truth

00–01

Ephemeris / Scenario
JPL DE442 kernels

Dynamics Propagation
Integrated state + STM

ZONE 2 · Vision / Measurement generation

04–05

Camera Model
Pinhole + FOV projection

Target Detection
Centroid in image plane

Pixel → Bearing
(u, v) → unit LOS

03

Bearing Measurement
Residual + covariance

ZONE 3 · Estimation

07

Phase 7 · Camera Pointing Logic
Repoints camera from filter estimate
so the target stays inside the FOV

CLOSED-LOOP NAVIGATION

ZONE 4 · Guidance / Correction

02

Apply Correction + Repropagate
Δv injected · truth updated

Midcourse Targeting Solve
Δv to reduce terminal miss

03

State Estimate
$\hat{x}$, P

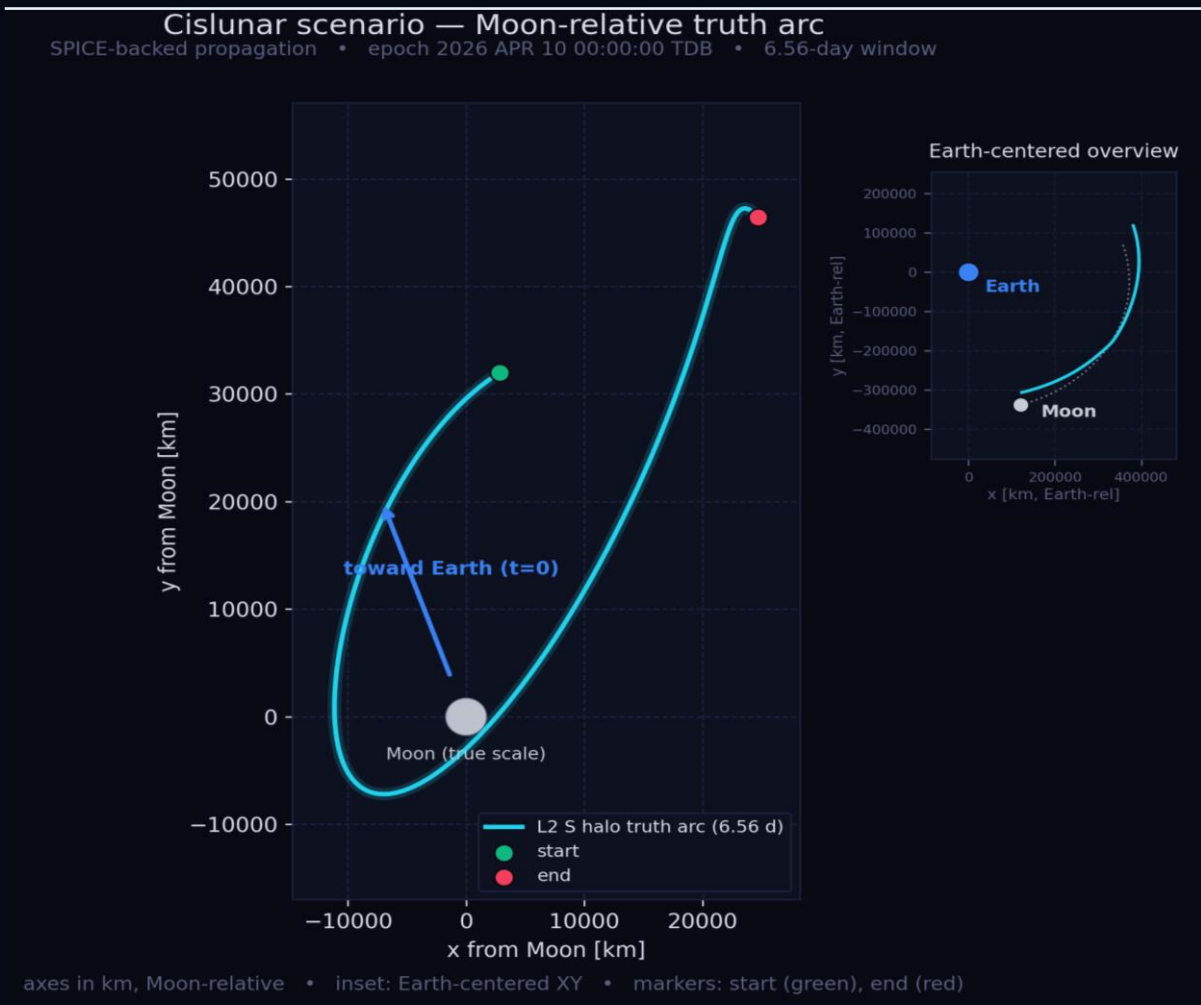
IEKF Measurement Update
Predict · Update (IEKF)

Without Phase 7
Fixed camera → target leaves FOV → measurements drop → estimator degrades

With Phase 7
Estimated LOS repoints camera → target stays in FOV → updates persist → stable filter

Cislunar Scenario: SPICE-Backed NRHO Propagation with Onboard LOS Geometry

The navigation pipeline is evaluated on a cislunar trajectory propagated from a periodic-orbit seed into a higher-fidelity Moon-relative scenario



Scenario

- Earth-Moon cislunar trajectory
- Moon-relative visualization
- SPICE-backed propagation used for higher-fidelity geometry

Propagation Model

- State: spacecraft position and velocity
- Dynamics: truth trajectory propagated in a cislunar grav environment
- State transition info used later by estimator + targeting solve

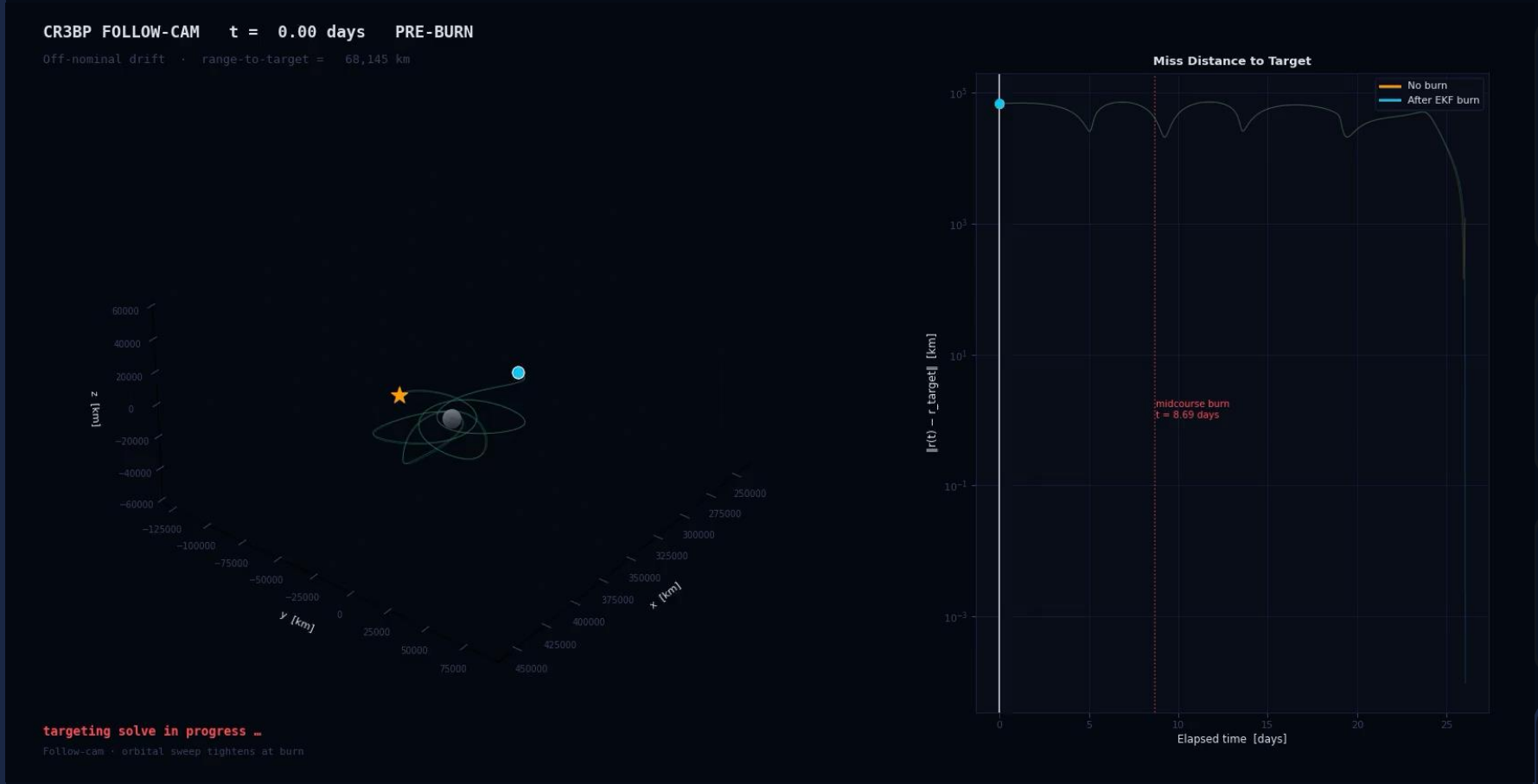
Importance of this setup

- Gives realistic geometry for LOS sensing
- Defines where burns and corrections are applied
- Lets filter and guidance results be interpreted in a mission-like context

Scope note: the scenario is a cislunar arc seeded from a periodic NRHO rather than a full TLI trajectory, so "transfer" in the proposal is exercised as a midcourse correction on a representative cislunar geometry.

Can the Estimated State Produce a *Useful* Correction Burn?

A single burn is applied at correction time t_c; the key question is whether the burn computed from the estimated state meaningfully reduces terminal miss



Correction Problem

- Choose a correction burn at time t_c
- Use the available state estimate at t_c
- Reduce final miss relative to the desired target state at the end of the arc

Three Cases

- No correction
- Perfect-information burn
- IEKF-based burn

Reported Metrics

- Terminal miss
- Burn error magnitude
- Relative Δv inflation

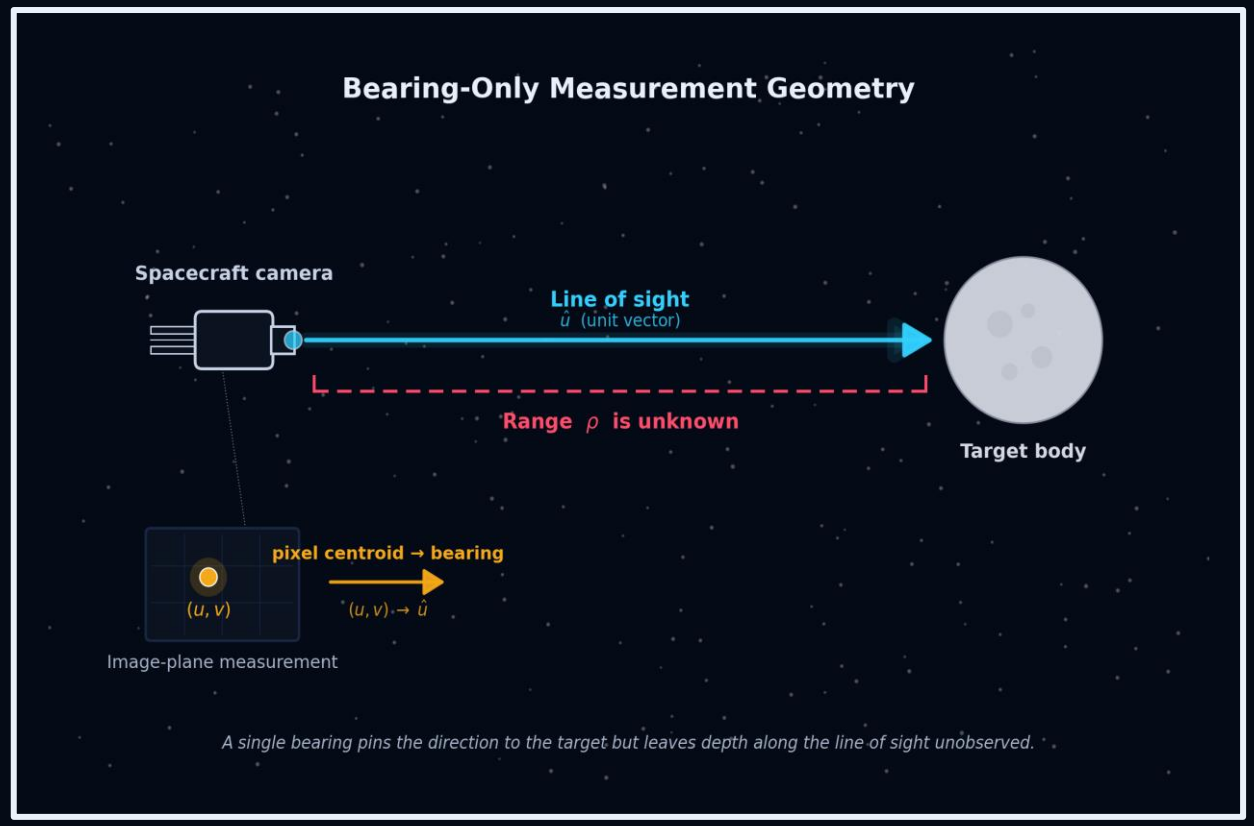
Definition Used Later:

$$Relative\ \Delta v\ inflation = \left(\frac{\|\Delta v_{IEKF}\|}{\|\Delta v_{perfect}\|} - 1\right) \times 100\%$$

Signed bias = |Δv_EKF| − |Δv_perfect|. Absolute error = ‖Δv_EKF − Δv_perfect‖.

A Camera Measures Line-of-Sight Direction, Not Range

The camera provides image-plane target location, which can be converted into a unit LOS vector, but the spacecraft-to-target distance is not directly observed



- **Measured directly:** image coordinates (u, v) in pixels
- **Converted measurement:** unit LOS vector $\boldsymbol{u}$
- **Not measured directly:** target range ρ

Bearing Measurement

$$\mathbf{u} = \left(\frac{\mathbf{r}_{body} - \mathbf{r}_{sc}}{\| \mathbf{r}_{body} - \mathbf{r}_{sc} \|} \right)$$

The observation constrains direction only

Why this is challenging

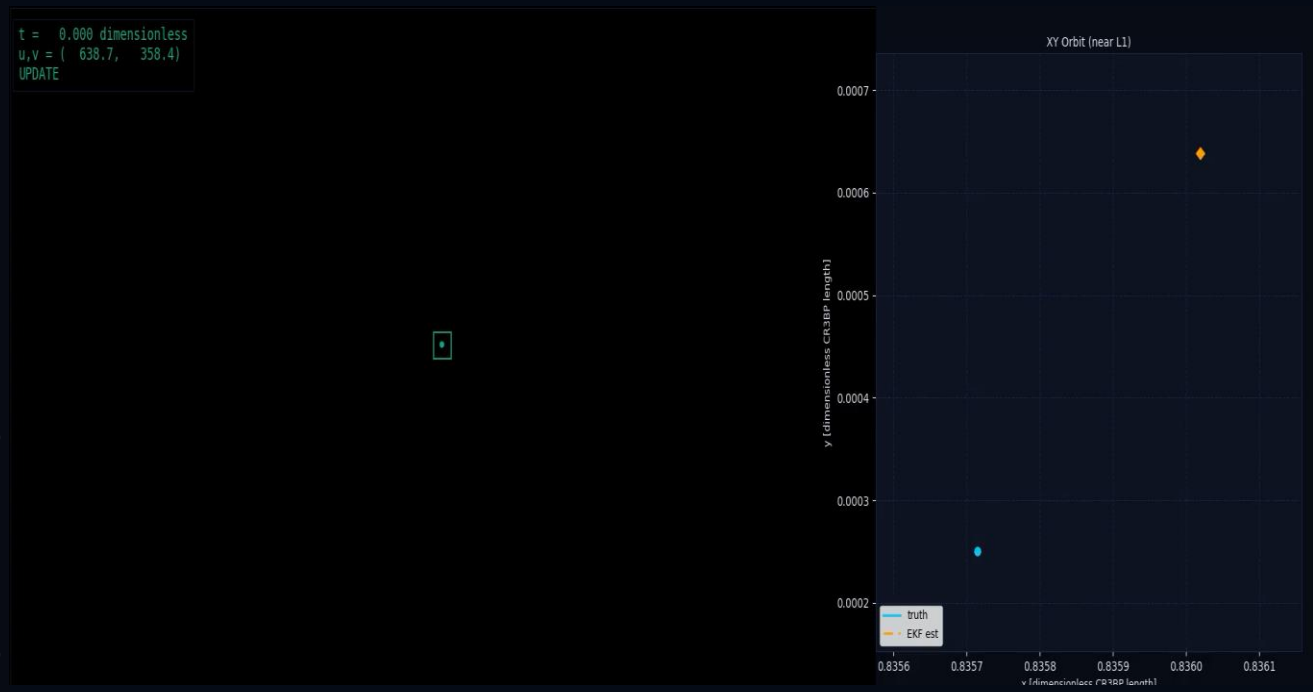
- Spacecraft state is 6-dimensional
- Bearing update is effectively 2-dimensional
- Distance must be recovered from dynamics + repeated geometry changes

Optical navigation is possible here because dynamics provide the missing structure that a single image does not

The Full Pixel-to-Bearing Pipeline Produces a Usable Navigation Estimate

The report summarizes estimation quality. The demo shows the target remains measurable on the image plane during the arc

Phase 05 Demo

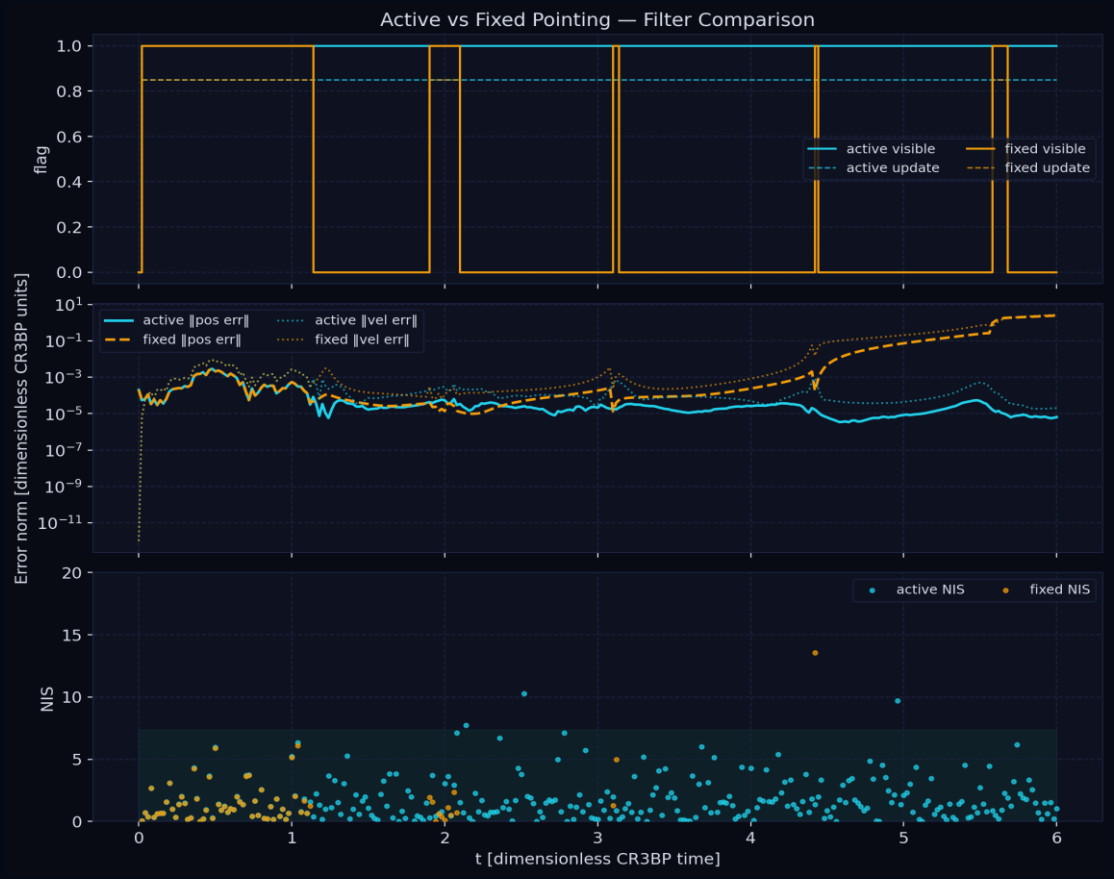


- Target remains measurable on the image plane
- Repeated pixel detections are converted to bearings
- Resulting estimate stays bounded and statistically interpretable

The pipeline depends on keeping the target measurable long enough for repeated pixel-to-bearing updates to improve the estimate

Keeping the Target in FOV Improves Measurement Availability

When the camera is repointed using the estimated LOS, the target remains visible longer, updates are sustained, and estimation quality improves relative to fixed pointing



What changes with active pointing?

- **Visibility increases:** target stays in FOV much longer!
- **Update rate increases:** more image measurements reach the IEKF
- **Error decreases:** active pointing maintains a much lower state-estimation error than fixed pointing

Metric	Fixed	Active
visibility	0.246	0.997
update rate	0.246	0.997
RMS pos error	5.4e-1	4.1e-4
RMS vel error	5.8e-1	1.4e-3

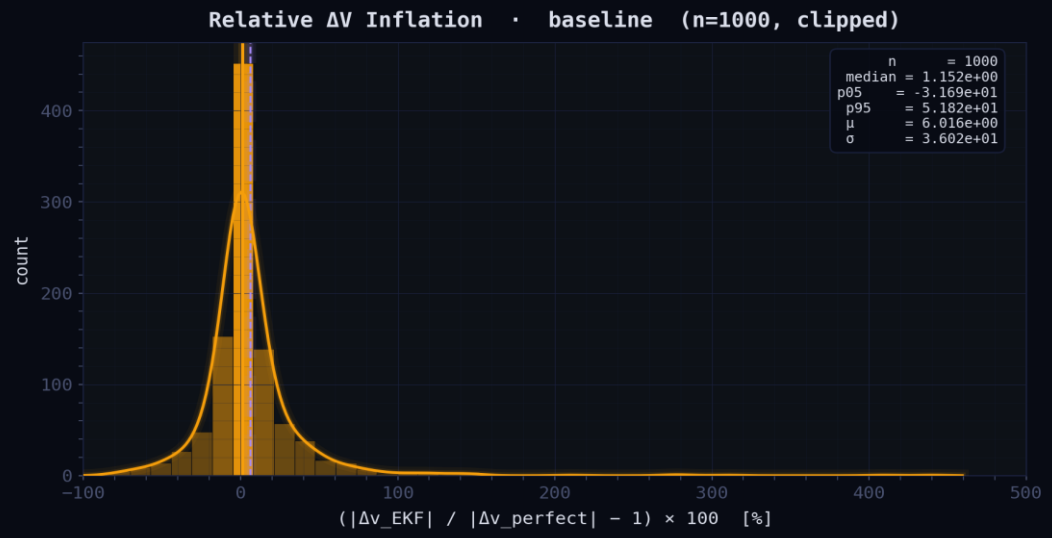
Active camera pointing becomes part of the navigation architecture because it preserves the measurements that the IEKF needs

Across Many Trials, IEKF-Based Correction Stays Near the Perfect-Information Benchmark

1000-run Monte Carlo cases compare the estimated-state burn against the perfect-information correction; reported values use dimensionless CR3BP units.

$$\text{Relative } \Delta v \text{ inflation} = \left(\frac{\|\Delta v_{IEKF}\|}{\|\Delta v_{perfect}\|} - 1\right) \times 100\%$$

A representative Monte Carlo trial shows the IEKF-based burn closely following the perfect-information correction geometry



Baseline MC · σ_px=1 · q=1e-14 · n=1000

|Δv_EKF−Δv_perfect| median = 2.91e-4, p95 = 8.15e-4

bias (|ΔvE|−|ΔvP|) median = +2.28e-5

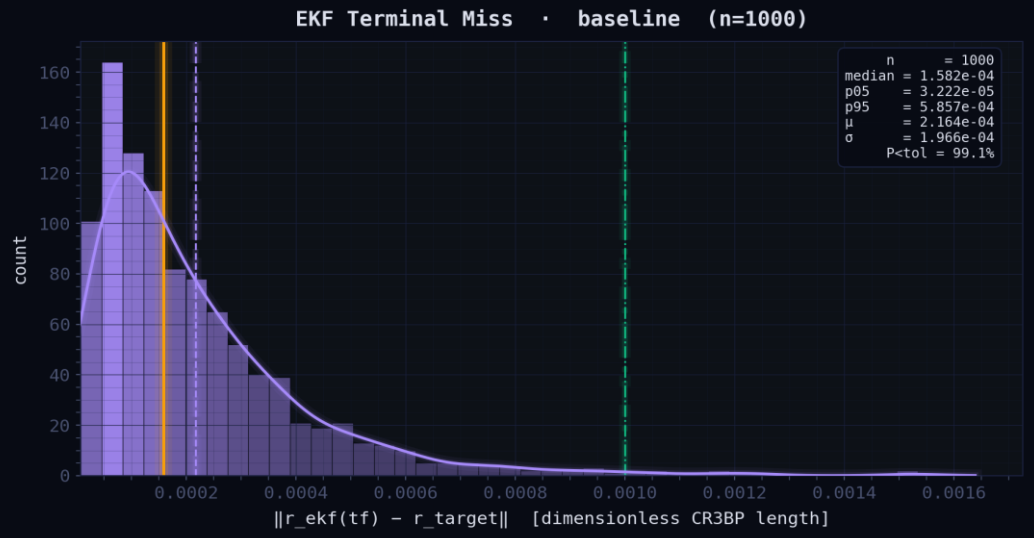
relative inflation median = 1.2%, p95 = 51.8%

terminal miss median = 1.58e-4, p95 = 5.86e-4 [ND]

success (tol=1e-3): 99.1%

High-noise stress · σ_px=5 · q=1e-6 · n=1000

miss median = 1.72e-3, p95 = 5.27e-3; success = 29.5%

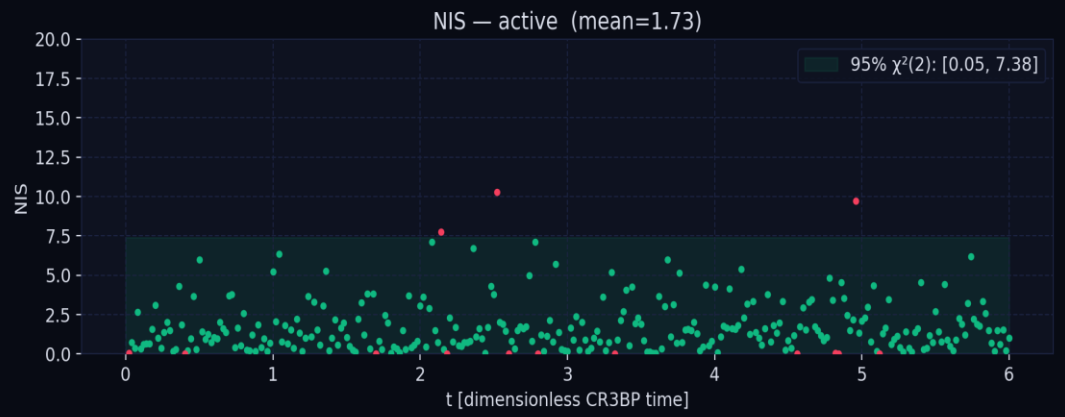


Across the baseline MC, the IEKF burn stays close to the perfect-information benchmark; the high-noise stress case shows the expected spread in burn quality.

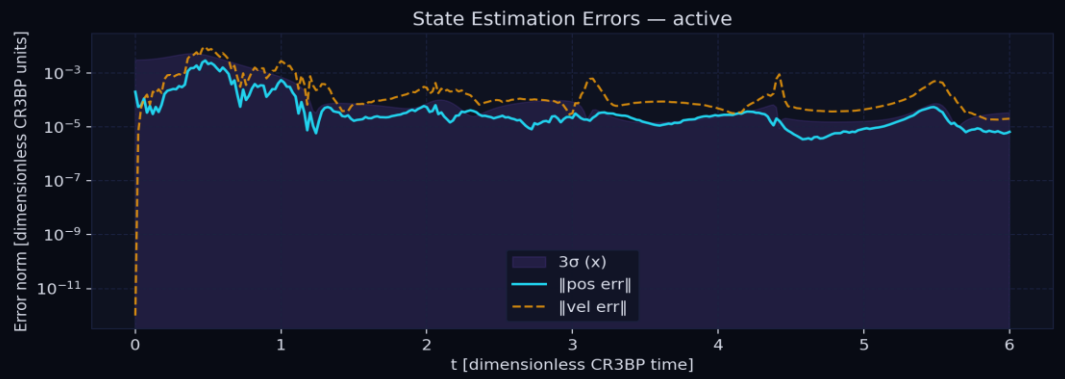
Filter Consistency: Does the IEKF Report Honest Uncertainty?

NIS, NEES, innovation whiteness, and 3σ position envelope on a representative run

4-panel diagnostic: NEES vs $\chi^2(6)$, P-trace decay, innovation ACF (whiteness), measurement-validity strip



NIS mean ≈ 1.73 vs $\chi^2(2)$ 95% band [0.05, 7.38] → IEKF innovations are consistent

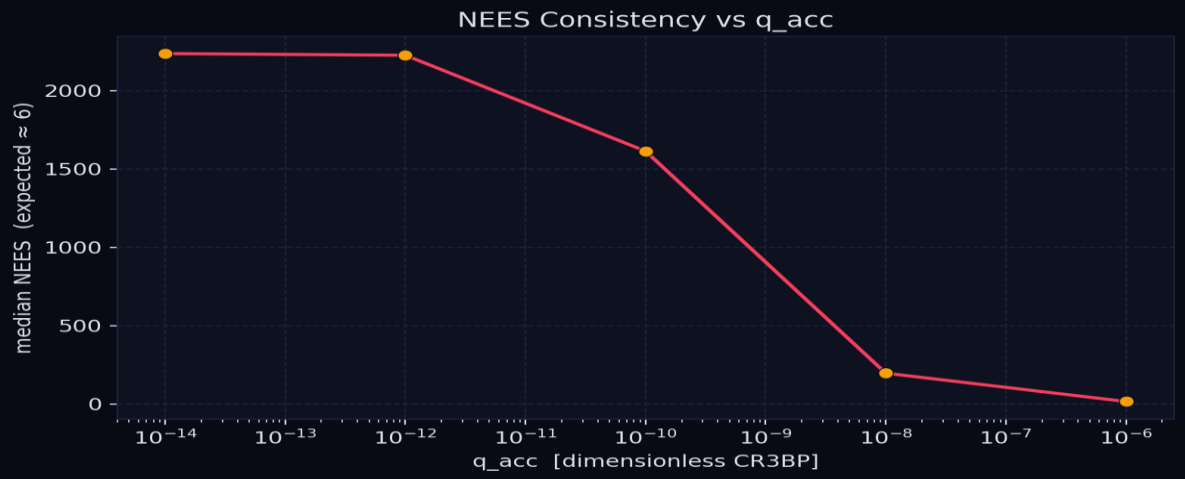


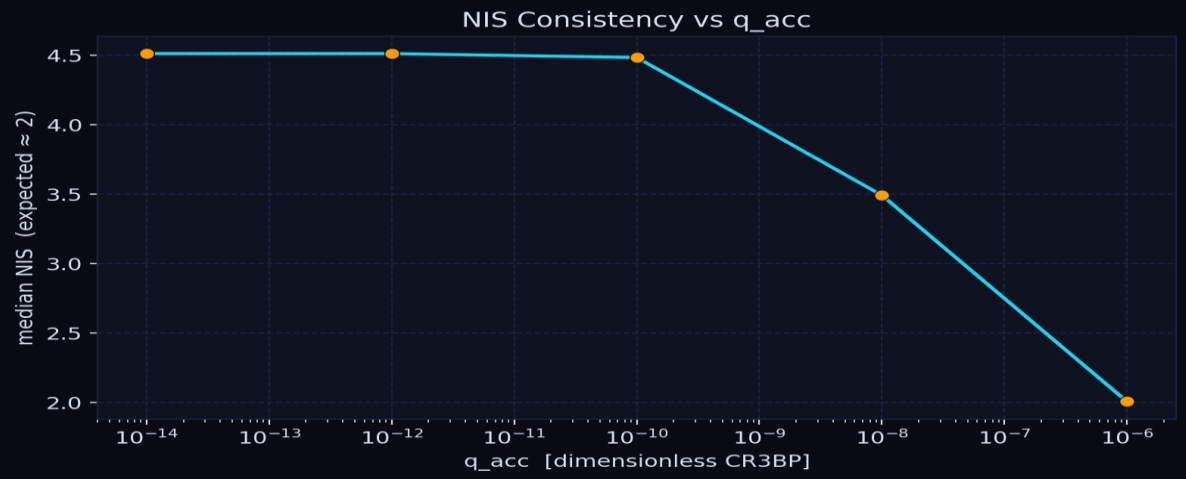
Position/velocity error hug the 3σ(x) envelope → covariance honest on the position-velocity channel

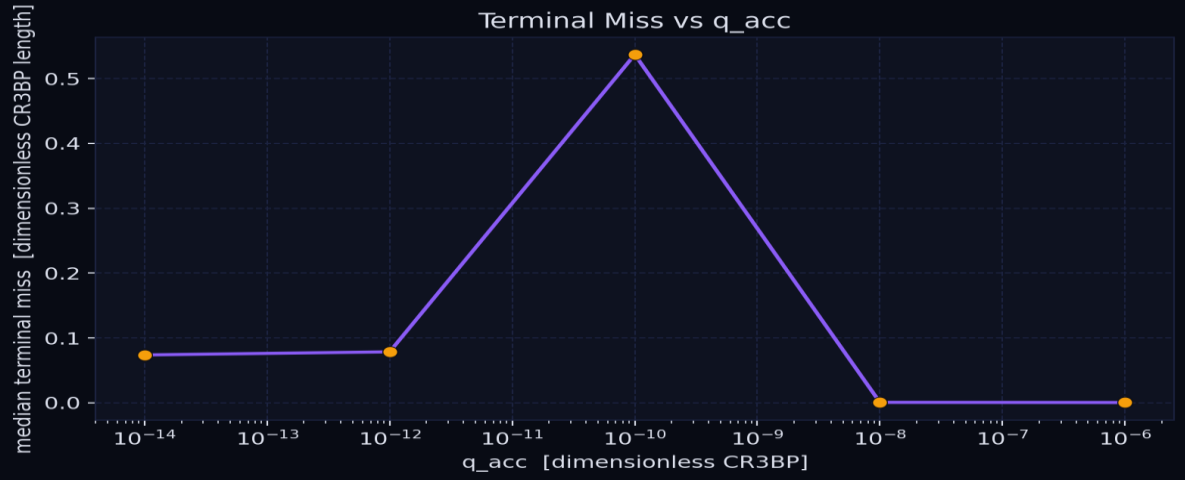
Finding: NIS and 3σ consistency look good on the 2-DoF bearing channel, but full-state NEES (6-DoF) is biased high; covariance propagation underestimates error growth. Motivation for the q_acc study

Process-Noise Tuning: A Bias-Variance Tradeoff

Sweeping q_acc from 1e-14 → 1e-6: baseline is overconfident; tuning buys consistency, costs median accuracy





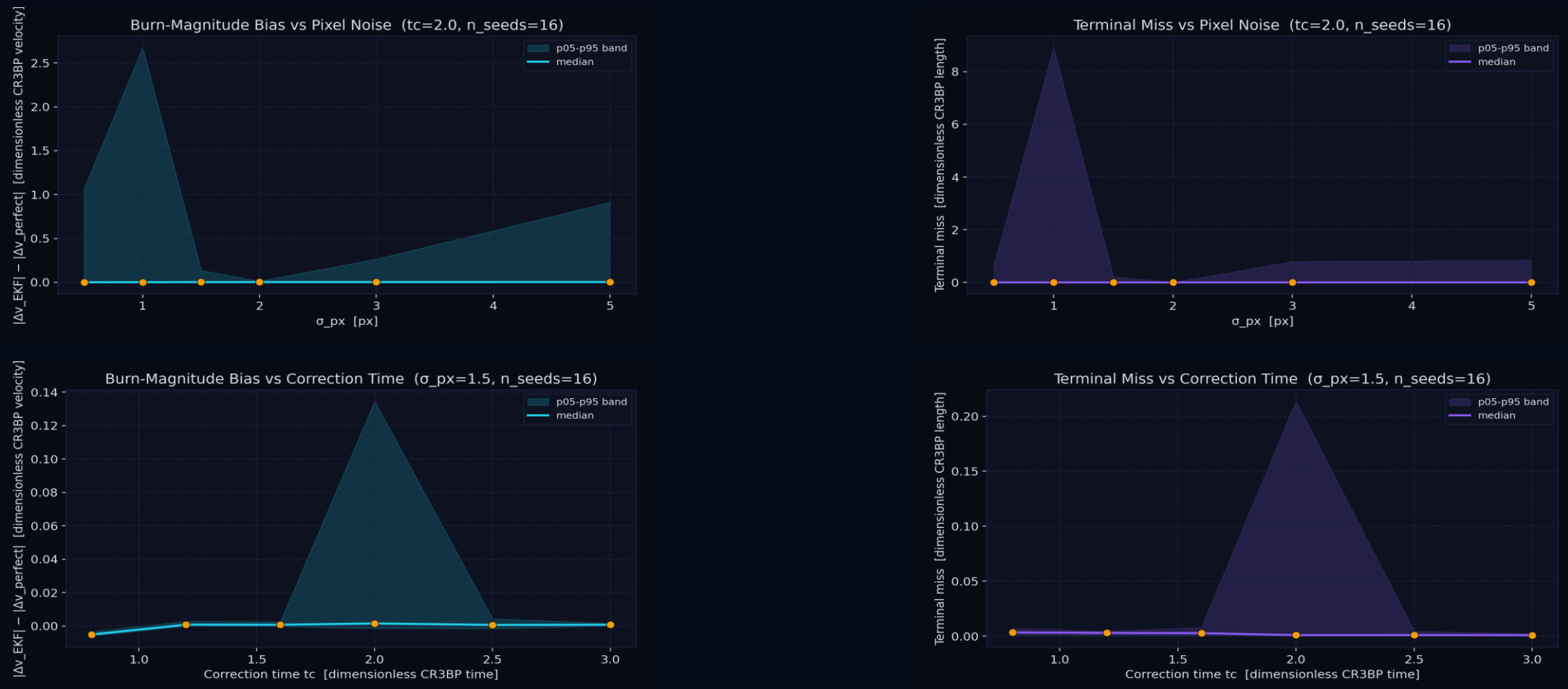


	Baseline q=1e-14	Tuned q=1e-6
NEES median	2237	15.2
NIS median	4.51	2.01
miss median [ND]	1.58e-4	7.19e-4
miss p95 [ND]	5.86e-4	2.00e-3

Takeaway: the baseline filter has tight point estimates but under-reports its uncertainty (NEES ≈ 300× expected). Raising q_acc restores χ^2 consistency at the cost of slightly wider burn distributions. The "right" q depends on whether downstream consumers need accurate covariance or tight median state.

Sensitivity Sweeps Show Where Bearing-Only Correction Becomes Fragile

Pixel noise and correction timing were swept while keeping the bearing-only IEKF and active pointing pipeline fixed.

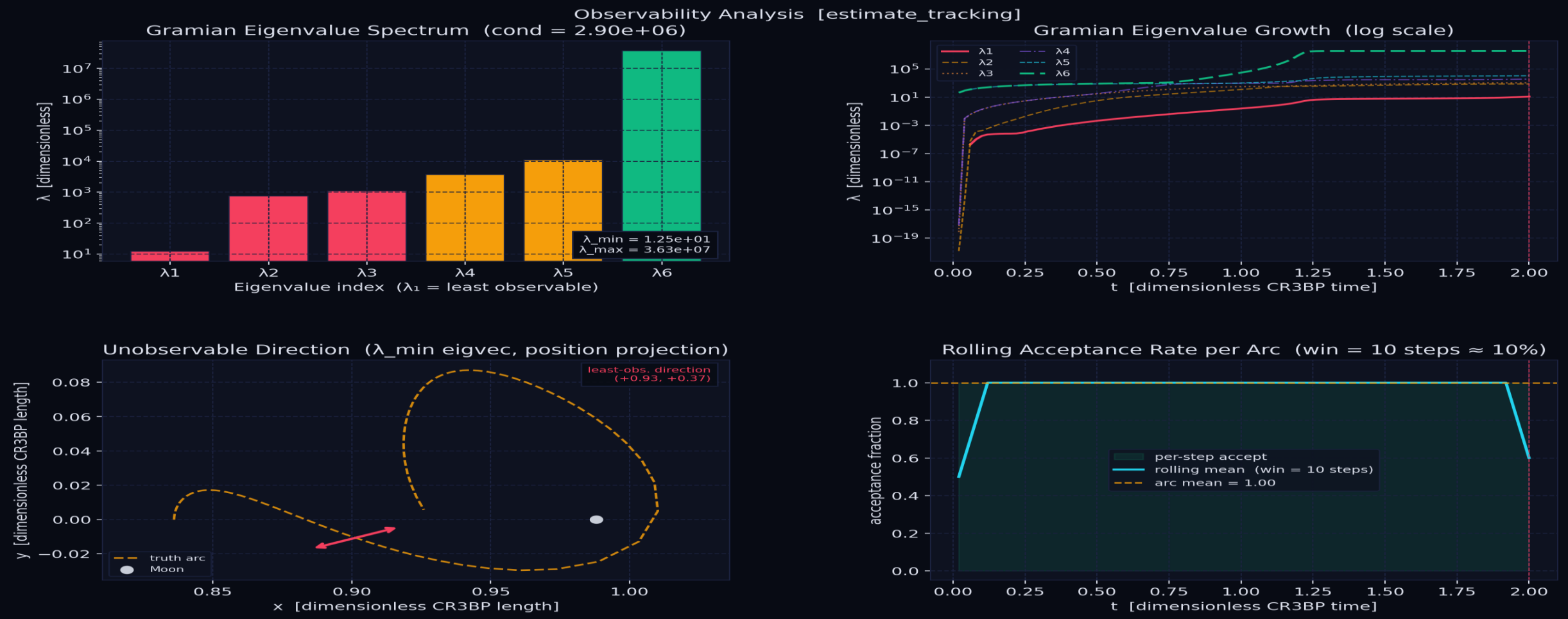


Multi-seed sweep (n=16 per grid point). Bands: p05–p95 across seeds; line: median. ND CR3BP units.

Key takeaway: estimation quality at the correction epoch, not just measurement availability, controls dV inflation and terminal miss.

Observability Gramian: Which State Directions Are Bearing-Observable?

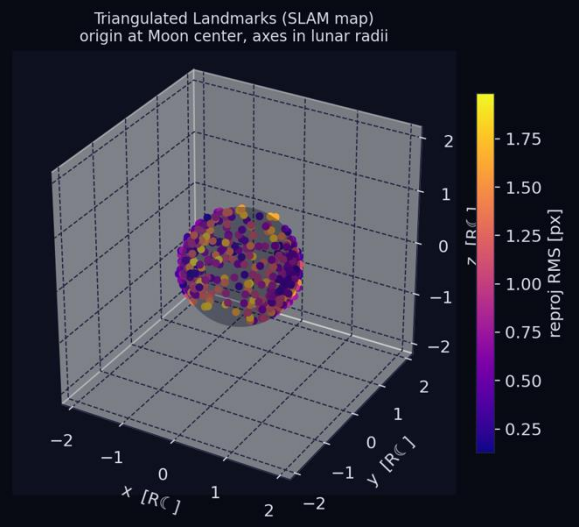
Accumulated Gramian $W = \Sigma\, \Phi^T H^T H\, \Phi$ — eigenvalue spectrum exposes the weakly-observable subspace

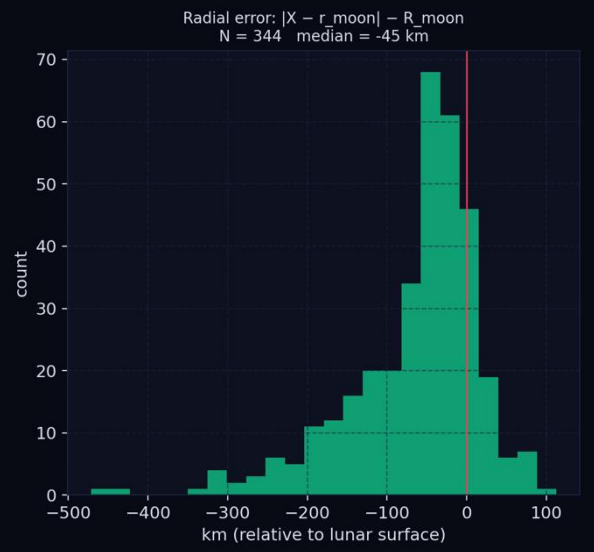


Eigenvalue spread ~6 orders of magnitude (cond(W) ≈ 3×10^6): position is strongly observable, but specific velocity directions (especially range-rate along the LOS) accumulate information slowly. Matches the Geller-Woffinden finding [13] that bearing-only cislunar nav needs either motion-induced parallax or a second observation target to cover range

[Future work] Feature Tracking Extends the Vision Pipeline Beyond a Single Centroid

Instead of reducing each frame to one image-plane centroid, Phase 08 tracks multiple visual features and triangulates candidate landmarks, opening a path toward richer optical navigation





What Phase 08 Demonstrates

Demonstrates: multi-feature tracking, candidate landmark geometry, and image-space diagnostics.
Not yet demonstrated: a fully integrated landmark-aided navigation filter.

Validation + Literature Comparison: What Is New?

The project is not claiming a new filter or the first optical-navigation result. The contribution is an end-to-end class-project pipeline that connects pixel-level bearing errors to cislunar correction-burn quality.

Reference / Study	Scenario	Measurement type	What they show	How this project differs
Bradley et al., 2020; Hinga & Williams, 2022	Cislunar optical state estimation	Natural/artificial targets or lunar landmarks	Optical angles can support cislunar state-accuracy studies	This work pushes from state estimation to correction-burn quality
Geller, Woffinden & Bieniawski, 2023	TLI-to-LOI robust TCM scheduling	OpNav passes with FOV / lighting constraints	TCM timing is sensitive to measurement availability and frequency	Here the sweep is pixel noise and correction epoch inside a nonlinear IEKF/MC loop
LONEStar and Artemis I OpNav, 2024	Flight optical-navigation demonstrations	Earth, Moon, planets, and starfield image processing	Real imagery validates the image-to-LOS OpNav measurement chain	This project closes simulated LOS measurements into an autonomous correction burn
This project	Cislunar correction arc	Pixel centroid -> bearing-only IEKF + active pointing	Estimated-state burn stays near perfect-info benchmark in baseline MC	Distinctive link: sensor accuracy -> state uncertainty -> dV inflation and terminal miss

Conclusions: Bearing-Only Optical Navigation Can Support Useful Cislunar Correction

This project evaluated whether pixel-based optical measurements, processed through an IEKF and active-pointing pipeline, can provide state knowledge good enough to support midcourse correction in cislunar geometry.

(1) A camera is a viable nav sensor when direct support is weak

In cislunar space, image-based sensing can provide useful onboard nav information when direct GPS-like support is limited, but the measurements are fundamentally bearing-only

Using pixel-to-bearing conversion and iterated filter updates, the pipeline produced bounded and decision-useful state estimates under repeated measurement availability.

(2) Bearing-only updates can recover a usable state estimate

(3) Camera pointing policy is part of the navigation architecture

Active repointing greatly improved visibility, update fraction, and estimation quality relative to fixed pointing, showing that keeping the target in view is a first-order systems issue

Across representative and Monte Carlo trials, the IEKF-based correction stayed close to the perfect-information benchmark and materially reduced terminal miss

(4) The estimated state supports useful correction

—

but the baseline covariance is overconfident (NEES ≈ 300× nominal); q_acc tuning restores χ^2 consistency at the cost of ~3× wider burn distributions.

The main result is not just that the filter runs. It is that a camera, an IEKF, and an active-pointing policy together form a usable cislunar optical-navigation pipeline

References and Resources

Core Texts

[1] Crassidis & Junkins, Optimal Estimation of Dynamic Systems, 2e, CRC, 2011
[2] Markley & Crassidis, Fundamentals of Spacecraft Attitude Determination and Control, Springer, 2014
[3] Christian & Robinson, Fundamentals of Spacecraft Optical Navigation, Wiley, 2022

Cislunar Optical Navigation (closest-to-this-work)

[4] Bradley & Olikara, "Cislunar Navigation Accuracy Using Optical Observations of Natural and Artificial Targets," J. Spacecraft & Rockets, 2020
[5] Christian, "A Tutorial on Horizon-Based Optical Navigation and Attitude Determination," IEEE Access, 2021
[6] Hinga & Williams, "Autonomous Cis-Lunar Navigation using Optical Measurements to a Lunar Landmark," NAVIGATION (ION), 2023
[7] Krause et al., "LONEStar: The Lunar Flashlight Optical Navigation Experiment," J. of the Astronautical Sciences, 2024
[8] Inman et al., "Artemis I Optical Navigation System Performance," AIAA SciTech, 2024
[9] Shimane, "Autonomous Horizon-Based Optical Navigation in High-Fidelity Cislunar Simulations," Georgia Tech SEAL, 2024
[10] NTRS 20210025728, "Robust Cislunar Trajectory Optimization via Midcourse Correction and Optical Navigation Scheduling," 2021
[11] "Robust Cislunar Navigation via LFT-Based H∞ Filtering with Bearing-Only Measurements," arXiv:2510.04942, 2025

Filter Consistency & Midcourse Correction

[12] Bar-Shalom, Li & Kirubarajan, Estimation with Applications to Tracking and Navigation, Wiley, 2001 (NIS/NEES χ^2 tests)
[13] Geller, Woffinden & Bieniawski, "Sensitivity of Optimal Midcourse Correction Scheduling to Navigation Performance," AAS GN&C, 2023
[14] autoNGC/CelNav on CAPSTONE, NTRS 20240014841, 2024

Project Code

GitHub repository: https://github.com/varshakrishnakumar/cislunar-optical-nav